completely escape all classes of human antibodies, and so, even if the immune system takes some time to recognise the virus, it does eventually destroy it, adds Dr Mishra.

THE NATURE OF THE BEAST

Like all viruses, the main aim of the Covid-causing virus, too, is to survive. That survival is predicated on two factors—the ability of the virus to continue infecting host cells to replicate, and then to spread to a new host to continue the process of replication. Currently, that survival is under threat due to a large number of people having acquired immunity. This is driving the virus to focus on mutations that will help it evade such immunity.

Evolutionary logic dictates that any variant of the Covid virus that allows it to continue spreading is a stronger, healthier strain. Currently, over 300 sub-strains of the dominant Omicron variant are jockeying to emerge as the fittest and most infectious. The WHO is closely monitoring all these sub-strains to see if any have mutations that are different enough to classify them as entirely new VoCs.

"We have not had a VoC for a while (since Delta was classified as one on May 31, 2021)," says Dr Jameel. "And it is no longer the case that there are significant jumps, rather there is a branching off from the same variant." Till early 2022, Covid displayed distinct 'jumps'—the Alpha, Beta, Delta, Gamma and Omicron variants had a cluster of mutations that gave each a competitive edge in spreading. However, these variants arose not from each other but from different Covid lineages. That changed with Omicron, with all new strains arising out of the parent variant. BA.1, BA.2, BA.3, BA.4 and BA.5 have each spawned sub-variants and recombinant strains, but Omicron remains the original variant. This behaviour is common with another major coronavirus—influenza. Each year, this virus, too, undergoes some mutations, but all out of one major strain rather than a fresh strain coming out of nowhere.

Alongside the tendency to splinter from the same branch, Covid has also shown a 'convergence' of mutations. "A virus doesn't have endless possibilities to mutate," says Dr Jameel. "This mutational space is reducing for Covid. Only those mutations that offer selective advantage to its ability to spread will get retained. And that is what we are seeing now—a sort of consolidation or convergence of mutations that offer Covid evolutionary advantages." One example is the L452R mutation, which was present in Delta and missing in the original but returned in its BA.5 sub-variant. Found in the spike protein of the virus, the mutation is widely associated with immune escape and stronger cell attachment, resulting in increased transmissibility and infectivity. "Covid is picking up immune escape mutations the most," says Dr Jameel, "as it allows it to continue to spread amongst populations."

Thus, if the original strain was estimated to be passed from 20 per cent infected people to others, Delta mutated to be twice as infectious, and Omicron turned out to be twice as infectious as Delta. Except that now the virus is finding very few people who do not have any immunity against it. Instead of causing waves of severe disease, therefore, the virus is triggering spikes of mild disease. "Data suggest all Omicron versions till date don't cause severe disease but continue to become better at evading immune responses and spread faster than earlier versions," says Dr Chauhan. Unlike Delta, which circulated in a previously unexposed population and caused hypoxia, lung disease and death, Omicron's major symptoms, per a 2022 *Lancet* study, are a predominantly sore throat, headache, runny nose, sneezing, cough and hoarseness of voice.

The result is, Covid has become no more lethal than seasonal influenza. The Spanish Flu of 1918 ended in two years after four major global waves of the disease. A strain of influenza, it went on to cause the Swine Flu and Russian Flu outbreaks as well. When a majority of the people gained immunity against it, the virus reduced in virulence and became seasonal in order to continue infecting smaller populations. Descendants



> **"No Omicron version till date has caused severe disease, but it continues to become better at evading immune responses and spread faster than earlier versions"**
>
> **— DR VIRANDER CHAUHAN**
> *Former director, ICGEB*